Alok Singh

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SKILLS

- **Machine Learning:** Regression, Classification, Clustering
- **Deep Learning:** CNN, RNN, Transformers
- **DevOps:** CI/CD, Docker
- **Languages**: Python, C++, C
- **Libraries/Frameworks**: NumPy, Pandas, Scikit-Learn, Seaborn, Matplotlib, PyTorch
- **Tools/Platforms**: VSCode, Jupyter Notebook, MySQL, GitHub, HuggingFace, Power BI, N8N
- **Soft Skills**: Quick Learner, Adaptability, Team Collaboration

INTERNSHIP

Freelancing – GPS Vehicle Tracking System (Work in Progress) Oct' 25 – Jan' 26

Self Employed

- Built a backend system using Node.js and MongoDB to handle real-time GPS tracking data.
- Processed device data through a TCP/UDP listener and stored it in MongoDB with live updates using WebSockets.
- Optimized database performance using indexing and caching for fast location tracking and alert generation.

  Tech: Node.js, MongoDB, WebSockets

PROJECTS

Neural Network for Cloudburst Prediction | *GitHub* June' 25 – July' 25

- Built a feedforward neural network from scratch using NumPy, achieving **87% prediction accuracy** on historical rainfall data.
- Reduced false positives by **18%** through threshold tuning and feature selection.
- Improved model performance by **22%** using outlier removal and normalization techniques.
- Developed a Flask web app serving **real-time predictions (<2s latency)** with interactive visualizations.

  Tech: Python, NumPy, Scikit-learn, Flask, Machine Learning

Airline Passenger Satisfaction Analysis | *GitHub* Apr' 25 – May' 25

- Designed an interactive Excel dashboard analyzing **100K+ passenger records** with dynamic KPIs.
- Reduced reporting time by **40%** using pivot tables, slicers, and automated visualizations.
- Identified key service factors impacting satisfaction, improving insight clarity by **30%**.
- Created **3 customizable dashboard templates** for stakeholder-ready presentations.

  Tech: Excel, Pivot Tables, Data Visualization, KPI Analysis

ACHIEVEMENTS

Top 10 Finalist (top -2%) WorldWide at the BRICS CCI Youth & AI Policy Lab - Case Study Challenge

- Analyzed the Shanghai 206 judicial AI system, identified severe data bias and poor explainability, and proposed an explainable AI (PSAM) approach to improve fairness and transparency in legal decision-making.

CERTIFICATES

- Youth & AI Policy Lab – Case Study Challenge | BRICS Chamber of Commerce & Industry Jan' 26
- Cloud Computing| NPTEL Nov' 25

EDUCATION

Lovely Professional University | Punjab, India Aug' 23 – Present

Bachelor of Technology - Computer Science and Engineering | **CGPA: 8.2**

Army Public School | Jodhpur, Rajasthan Apr' 21 – Mar' 22

Intermediate | **Percentage: 83**

Army Public School | Jodhpur, Rajasthan Apr' 19 – Mar' 20

Matriculation | **Percentage: 78**